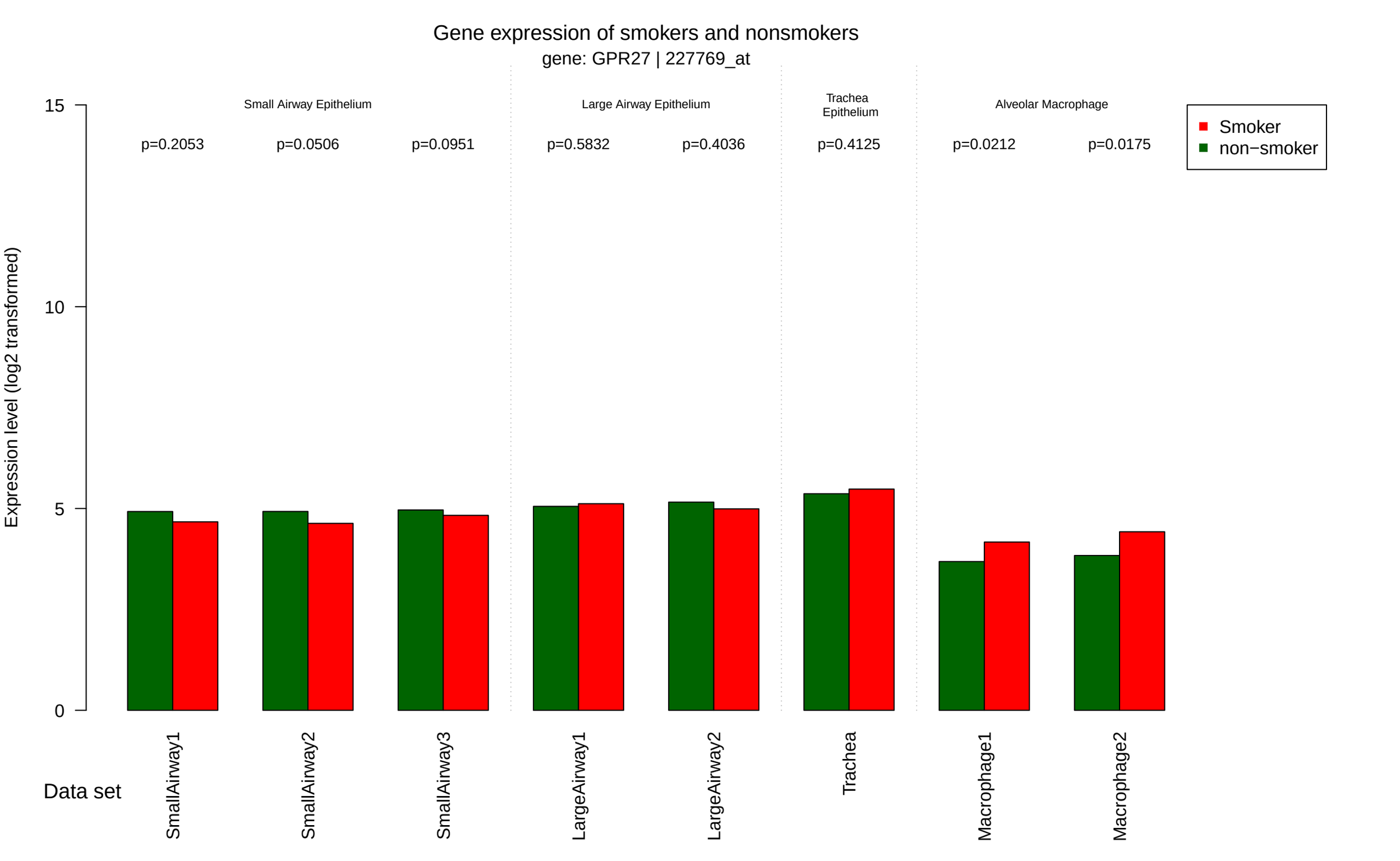
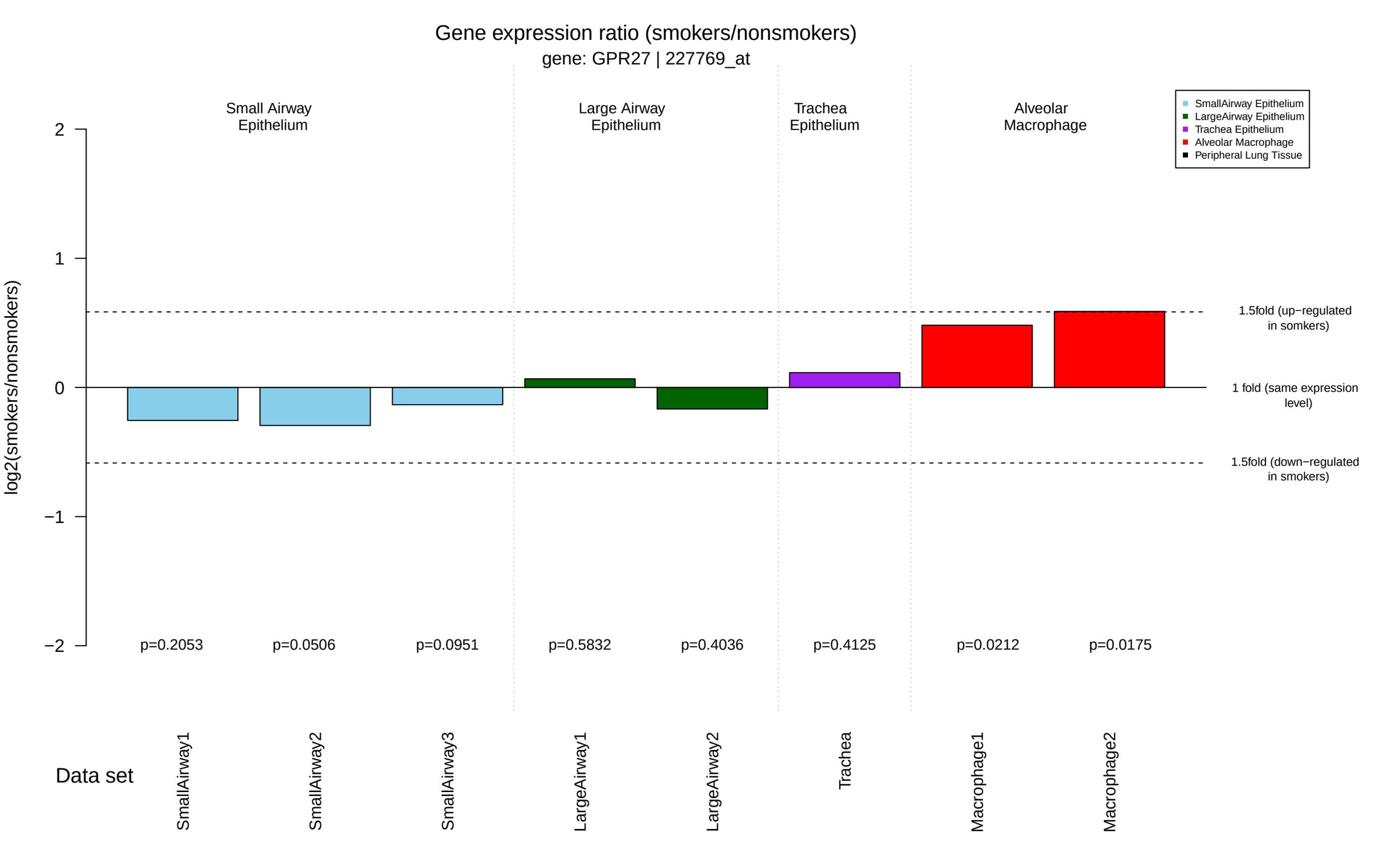
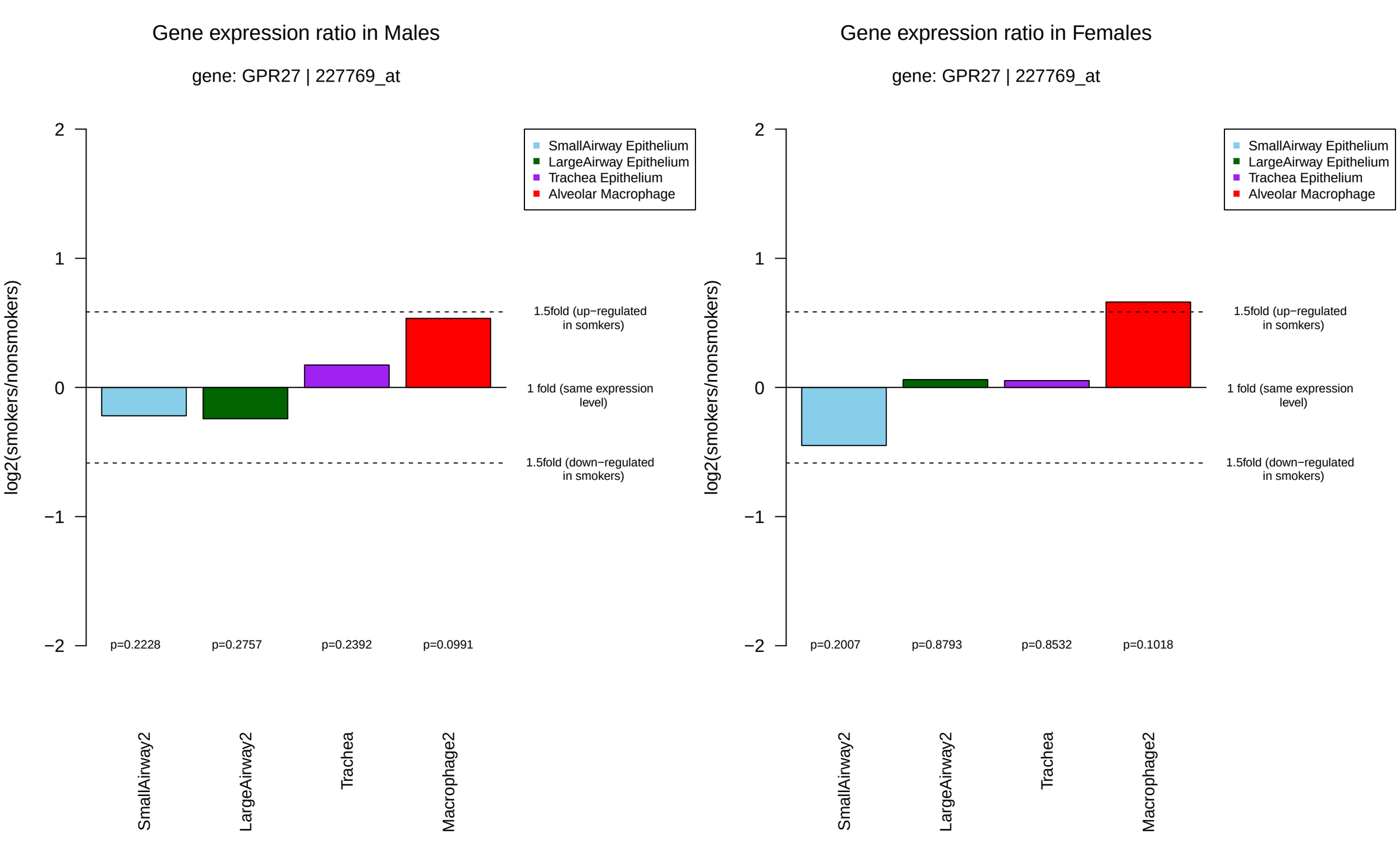
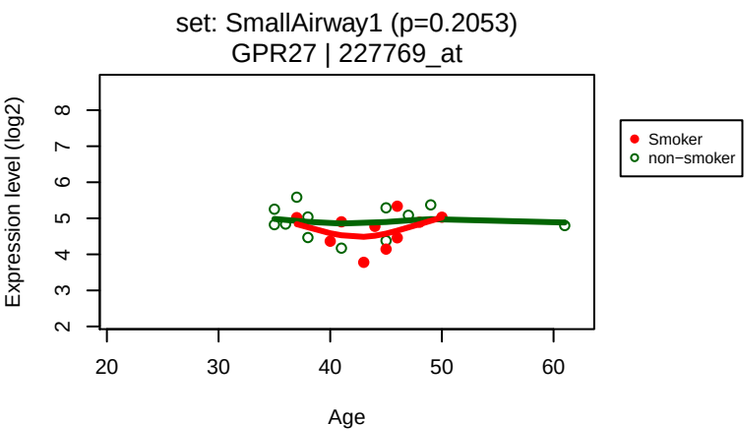
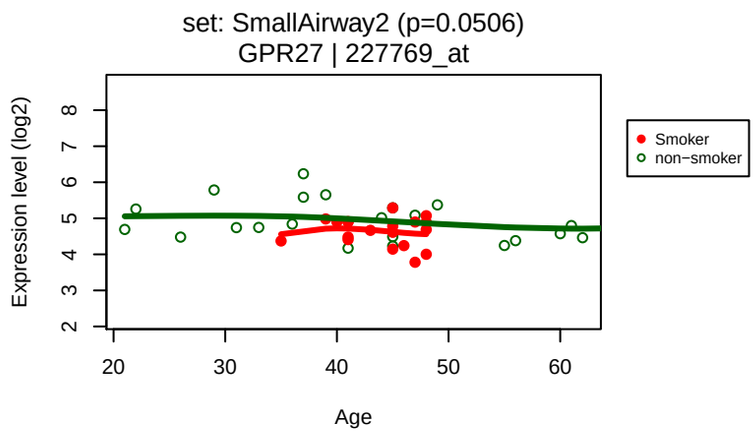
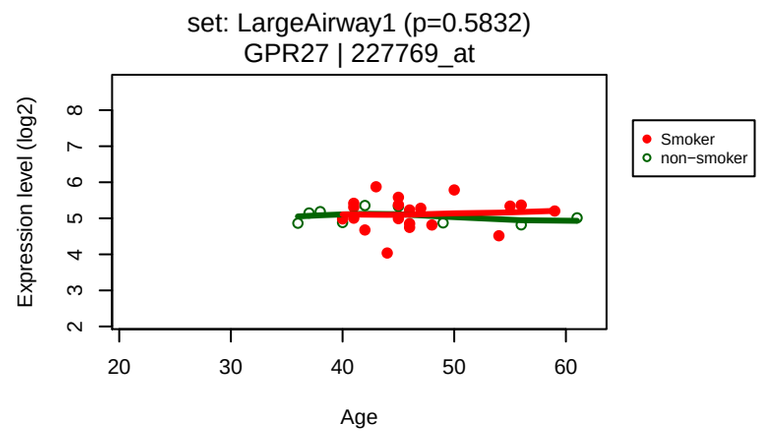

Gene expression of smokers and nonsmokers
gene: GPR27 | 227769_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2053
p=0.0506
p=0.0951
p=0.5832
p=0.4036
p=0.4125
p=0.0212
p=0.0175
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

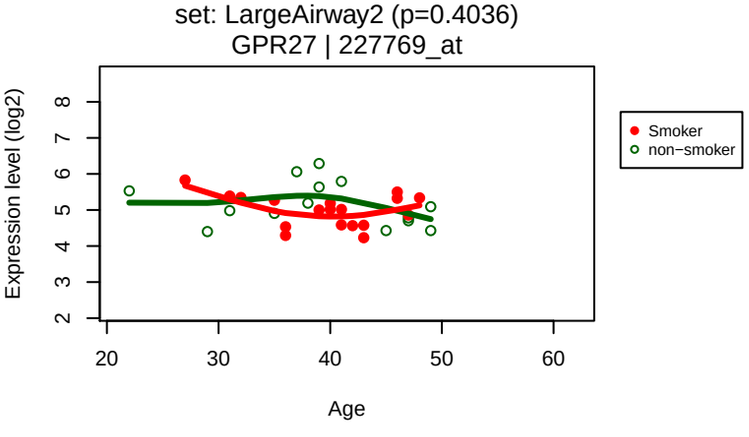

Gene expression ratio (smokers/nonsmokers)
gene: GPR27 | 227769_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2053
p=0.0506
p=0.0951
p=0.5832
p=0.4036
p=0.4125
p=0.0212
p=0.0175
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

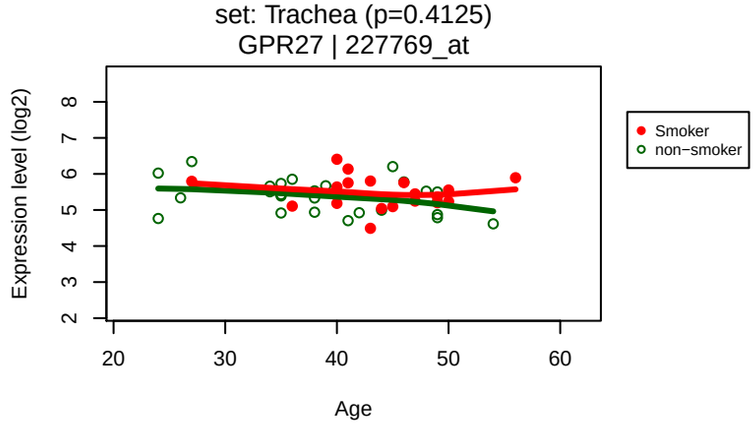

Gene expression ratio in Males
gene: GPR27 | 227769_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2228
p=0.2757
p=0.2392
p=0.0991
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR27 | 227769_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2007
p=0.8793
p=0.8532
p=0.1018
SmallAirway2
LargeAirway2
Trachea
Macrophage2

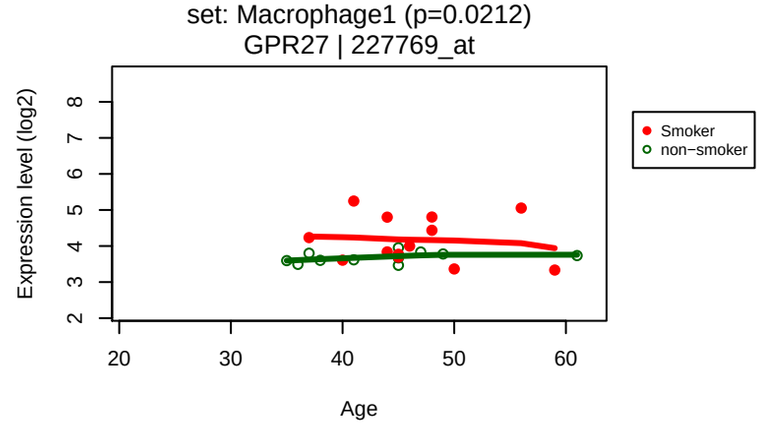

set: SmallAirway1 (p=0.2053)
GPR27 | 227769_at
Expression level (log2)
Age
Smoker
non−smoker

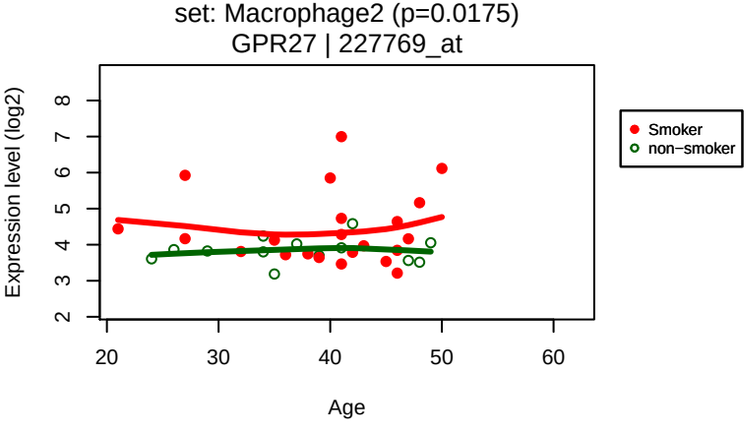

set: SmallAirway2 (p=0.0506)
GPR27 | 227769_at
Expression level (log2)
Age
Smoker
non−smoker

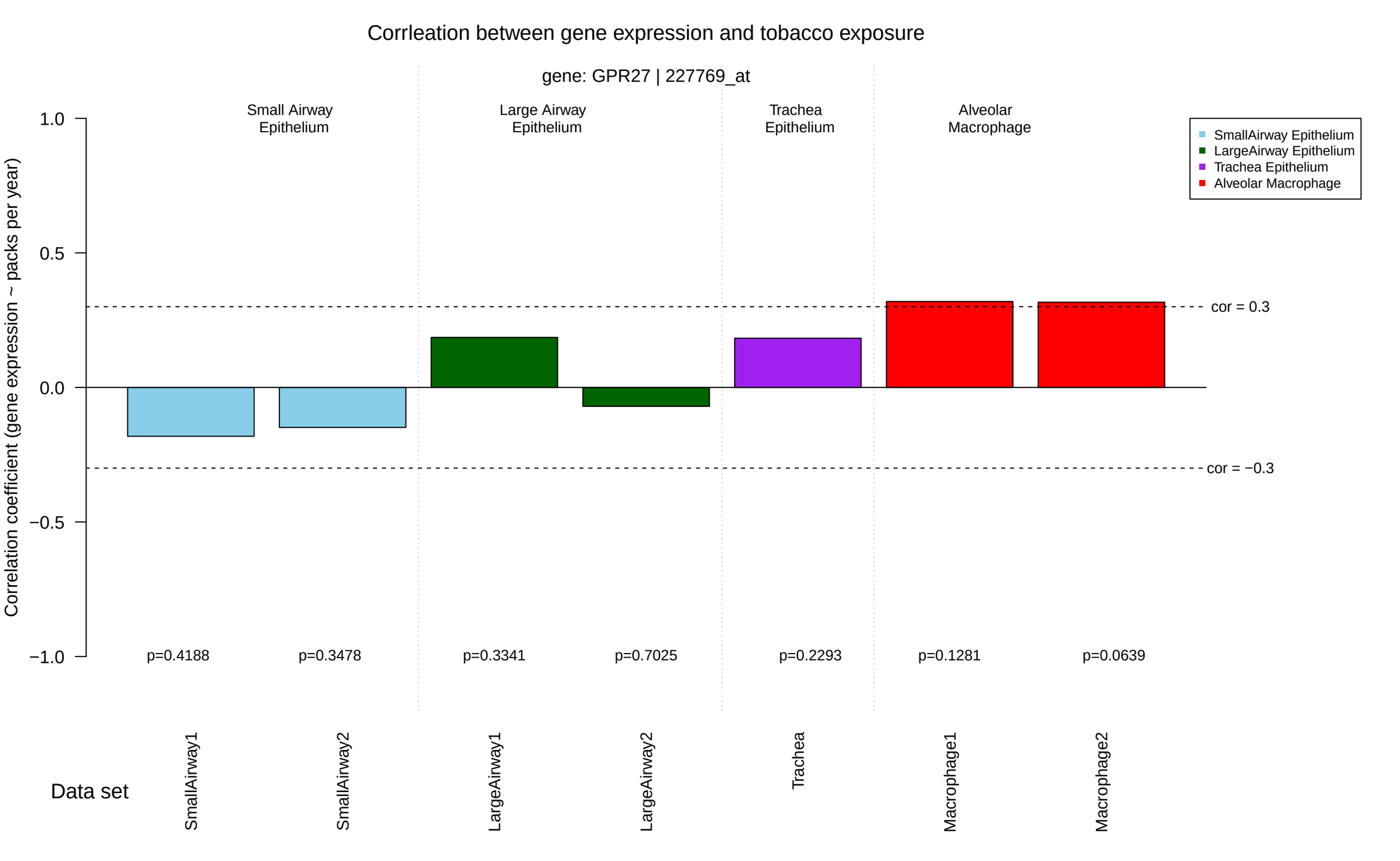

set: LargeAirway1 (p=0.5832)
GPR27 | 227769_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4036)
GPR27 | 227769_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4125)
GPR27 | 227769_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0212)
GPR27 | 227769_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0175)
GPR27 | 227769_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR27 | 227769_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4188
p=0.3478
p=0.3341
p=0.7025
p=0.2293
p=0.1281
p=0.0639
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2